

Validating Efficiency of the Forward-Forward Algorithm

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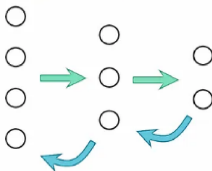
- ▶ Perceptron introduced in late 1950s
- ▶ Backpropagation (BP) algorithm developed in 1986
- ▶ BP as the fundamental method for training MLP, CNN, GAN, large language models (LLM)
- ▶ Strength: fine-tuning network parameters through iterative process

- ▶ Scalability issues with growing model size and complexity
- ▶ High computational power and memory requirements
- ▶ Biological implausibility
 - Human brain shows no evidence of backpropagating information
- ▶ Environmental concerns

Forward-Forward Algorithm by Geoffrey Hinton

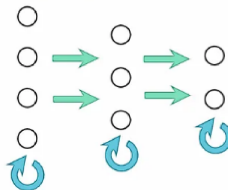
Backpropagation

Forward → Backward



Forward-Forward

Forward Forward → Local update



Nebuly^{AI}

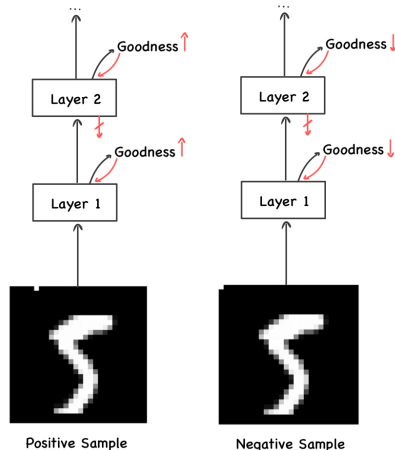
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- ▶ Motivated by how the human brain works
- ▶ Key difference: two forward passes **instead** of forward-backward cycle

The Forward-Forward Algorithm (cont.)



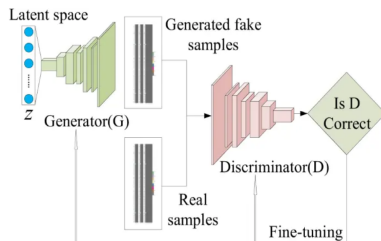
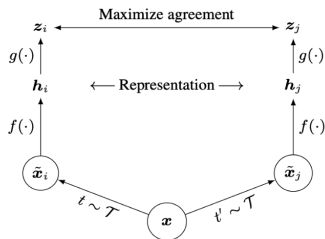
- ▶ How it works:
 - Positive pass - real data
 - Negative pass - fake/random or generated data
- ▶ Objective
 - Positive pass - **Maximise** *goodness*
 - Negative pass - **Minimise** *goodness*
- ▶ Goodness - Sum of Squared (neural) activities
 - Other functions not in scope of project



from:

<https://github.com/loewex/forward-forward>

► Forward-Forward inspirations



What's GAN (generative adversarial networks), how it works?

► SimCLR: a Simple framework for Contrastive Learning of Visual Representation

► GAN: Generative Adversarial Networks

► Layer-wise

$$p(\text{positive}) = \sigma\left(\sum_j y_j^2 - \theta\right)$$

- σ - logistic function
- θ - threshold (*hyperparameter*)
- y_j - ReLU activation in layer j before normalisation

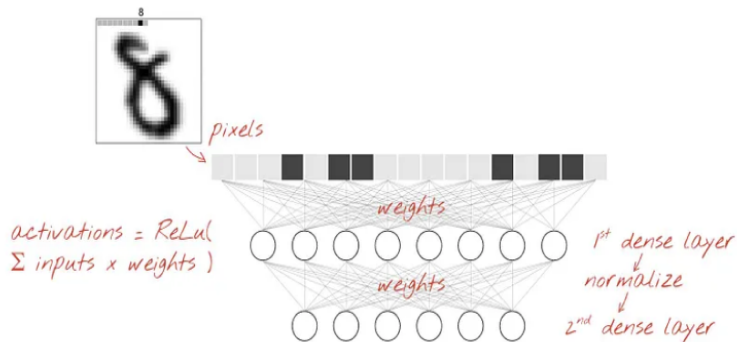
► Goal:

- High values for positive data
- Low values for negative data

The Forward-Forward Algorithm (cont.)



Layer Normalisation:



► Normalisation

- Prevent subsequent layers from cheating
- Only vector orientation, not length is passed onto subsequent layers

- Potential advantages:

- Faster training time
- No storing of neural activities = less resource intensive
- Offline parallel computing with *negative* data

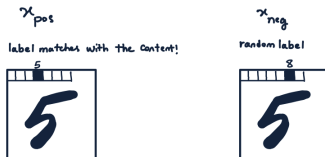
- ▶ Early research stage with various applications being explored

¹<https://primo.ai/index.php?title=Forward-Forward>

Does the Forward-Forward algorithm offer significant improvements in training time and/or memory efficiency compared to backpropagation for supervised classification² tasks?

²Section 3.3 of Hinton's paper

- ▶ Two key outcomes:
 1. Better understanding of the forward-forward algorithm
 - ▶ practical neural network implementation
 2. Determine if there are significant efficiency advantages (FF vs. BP)
 - ▶ if not, what are the trade-off(s)?
- ▶ Contribute to development of novel techniques for training neural networks



3

► MNIST Dataset

- 60,000 hand written digits.
- 28×28 pixel grey-scale images.

► Forward-Forward:

- Class labels one-hot encoded and appended to each image
- Negative samples generated by randomly assigning class label.

³Sourced from https://github.com/mpezeshki/pytorch_forward_forward

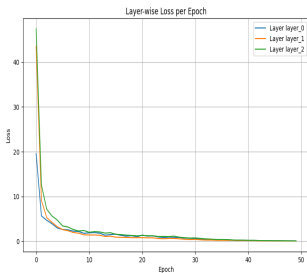
Experiment approach

- ▶ Epoch Control
- ▶ Test Accuracy Control

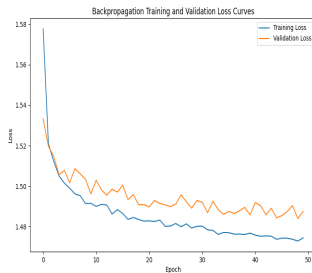
Build two neural networks:

- ▶ One using FF ($\theta = 2$) and the other using standard BP
- ▶ Identical, fully-connected network
 - 3 hidden layers
 - 1000 ReLU neurons per layer.

1. Train each model for 50 epochs.
2. Plot trained parameters - loss curves, t-SNE, weight distribution
3. Record the following metrics per epoch:
 - 3.1 GPU utilisation (%)
 - 3.2 Power Usage (Watts)
 - 3.3 GPU Memory (Mb)
4. Calculate the ratio between FF and BP for each metric across epochs.
5. Bootstrap values to determine there is statistically significant difference.

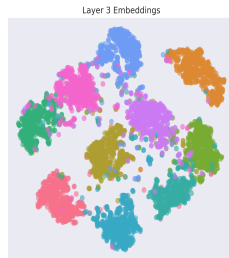
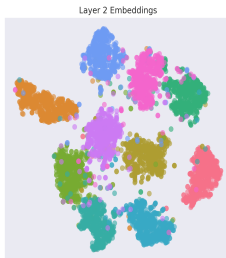
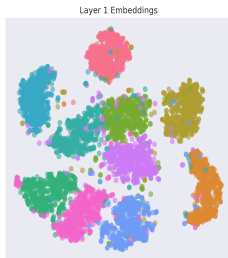
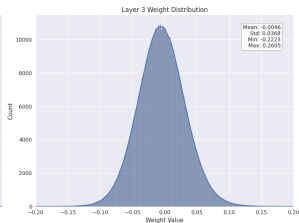
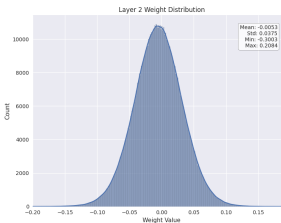
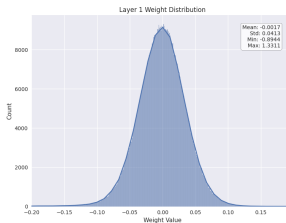


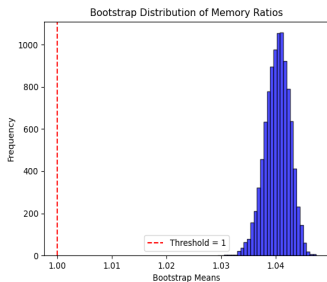
(a) Forward Forward



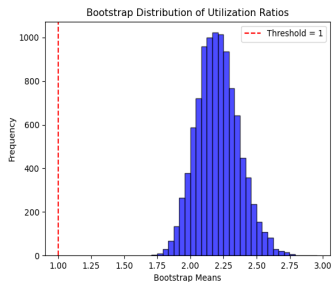
(b) Back propagation

Figure 2: FF vs BP loss curves





(a) Memory usage ratio



(b) Memory utilization ratio

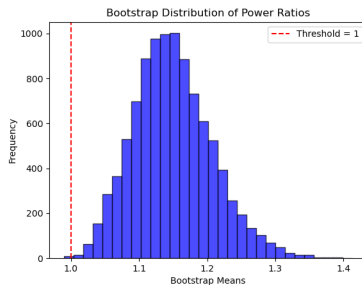


Figure 4: Power Usage Ratio

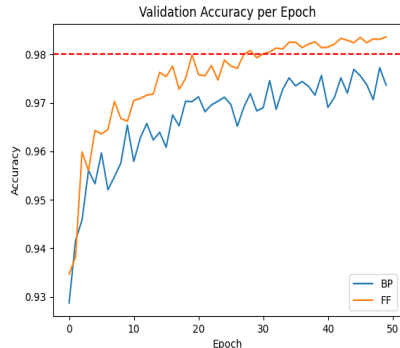


- ▶ FF performed worse on per epoch basis.
- ▶ FF took longer overall.
- ▶ Decided to use test accuracy as control.

Approach #2 - Test Accuracy Control



1. Train until 97% test accuracy.
2. Record at frequent time intervals:
 - GPU utilisation (%)
 - Power Usage (Watts)
 - GPU Memory (Mb)
3. Repeat the above step 10 times.
4. Calculate average values and standard deviation.



Metric	BP		FF	
	Mean	Std	Mean	Std
Runtime (sec)	268.29	0.70	218.07	5.58
Power (watts)	19.36	0.60	19.41	2.63
Total Power Used (Joules)	5193.06	158.95	4238.42	621.09
Utilization (%)	36.03	0.49	27.39	1.60
Memory (MB)	1462.98	9.53	1466.69	6.03
Energy (kWh)	0.001442	-	0.001177	-

Table 1: Model metrics comparison at threshold accuracy (97%) over 10 runs



- ▶ FF reaches test accuracy faster.
- ▶ FF utilizes the GPU less.
- ▶ Uses approximately 20% less power.



- ▶ Experiment conducted on a gaming GPU instead of HPC
- ▶ Experiment was only conducted 10 times
- ▶ Not enough data to do statistical test for significance

- ▶ Confirms Hinton's hypothesis that FF is a more efficient algorithm
- ▶ More suited for limited resource computing.
 - Embedded "on-chip" devices
 - IoT systems
- ▶ BP will likely remain the dominant algorithm.

- ▶ The models were run on gaming GPU.
- ▶ Back propagation code can be optimized.
- ▶ Statistical significance of test accuracy control results over more iteration samples
- ▶ GPU monitoring code affected model performance.

- ▶ Use different datasets.
- ▶ Different network sizes.
- ▶ Different types of neural network.
- ▶ Parallelisation.

- ▶ Forward-Forward algorithm - promising new method for more efficient NN training.
- ▶ Novel approach - positive/negative data, local updates
- ▶ Works well in small-scale tasks
- ▶ Experiments confirms efficiency in power usage & GPU utilisation, but additional testing required
- ▶ BP will likely remain the dominant algorithm.
- ▶ FF may find a niche in embedded and IOT devices.

Thank you! Any Questions?